

A shock too steep to reflect cleanly: how well do we track it?

A Mach 10 shock meets a wall at 60° and cannot reflect off it attached, so it lifts away and a second shock forms underneath to carry the turn. Two-dimensional, inviscid, integrated to $t = 0.2$ on two grids.

The answer in one line

One quantity in this problem is known exactly: the incident shock travels at a speed the shock relations fix, so its position at any instant is a closed form. Solved, it sits within **0.15 %** of the distance it travelled on the fine grid and 0.17 % on the coarse one, against a criterion of 1 % that was written down before the first mesh was built. The finer grid landed closer than the coarser one, which is what was predicted in advance. Both differences are smaller than one cell of their own grid.

Table 1 – Incident shock position at $t = 0.2$, against the exact solution

Grid	Cells	Spacing	Solved position	Exact position	Difference	Share of distance travelled [%]	Resolution of the measurement [-]
	[-]	[-]	[-]	[-]	[-]		
Coarse	14,400	0.016667	3.00448	3.00049	3.98691×10^{-3}	0.17%	± 0.01667
Fine	57,600	0.008333	2.99666	2.99328	3.38139×10^{-3}	0.15%	± 0.00833

Lengths are in the problem’s own non-dimensional units, in which the wall runs from 0 to 4 and the domain is one unit tall. The distance the shock travelled between the start and $t = 0.2$ at a fixed height is $20t/\sqrt{3} = 2.30940$, and the sixth column is the difference as a share of it. The final column is the spacing at which the shock position can be resolved on that grid, which is the grid spacing itself.

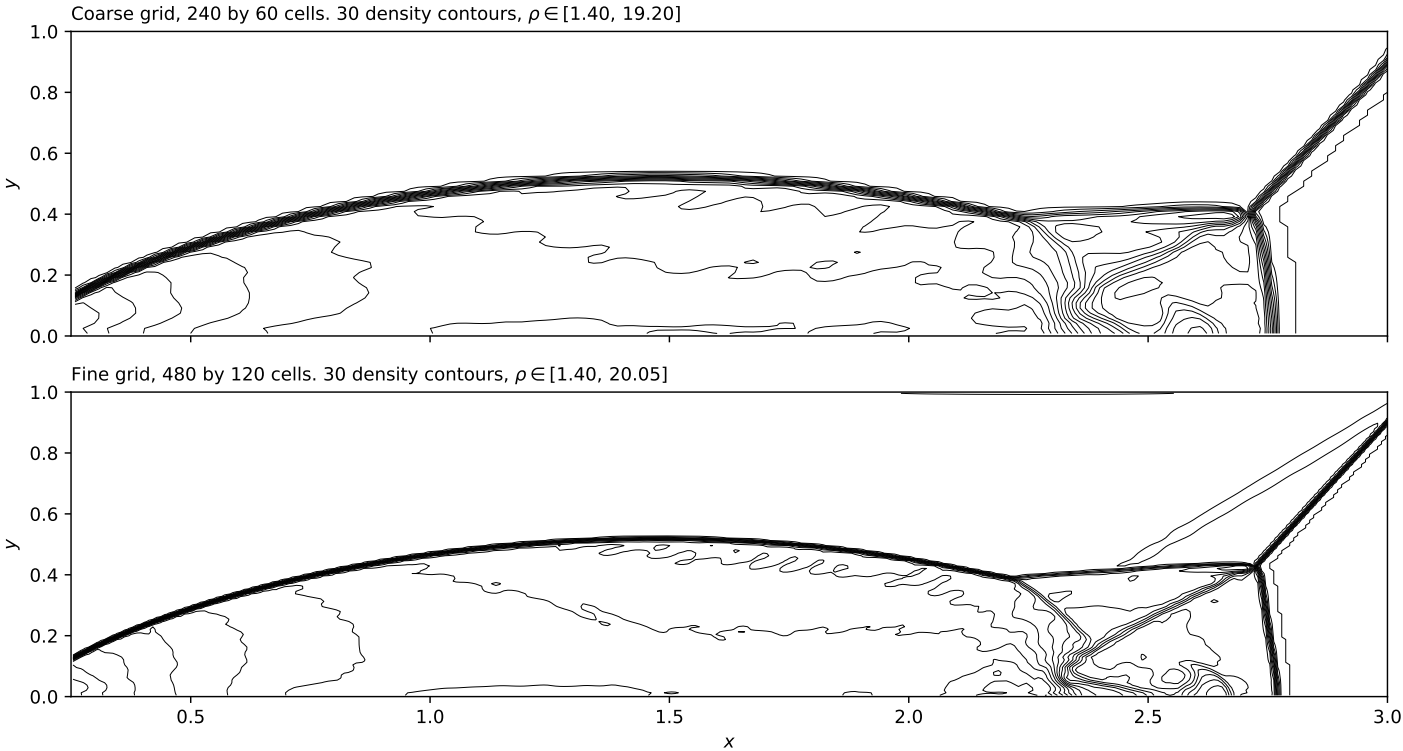
Table 2 – Compute

Item	Core-minutes	Basis
Set aside before running	20	fixed with the success criteria
Used	2.4	both grids, meshing and post-processing included
Used as a share of what was set aside	0.12	—

A core-minute is one processor working for one minute; all work here ran on four at once. Cost in currency is not quoted because this machine cannot read its own billing.

Density at $t = 0.2$

Double Mach reflection: density at $t = 0.2$, two grid resolutions



The main shock arcs from the lower left. Where it leaves the wall, a second shock and a jet along the wall form beneath it. The fine grid holds those features tighter than the coarse grid does.

What was verified, in plain words

Lead Numericist. The incident shock position was compared against the exact closed-form solution for the shock speed at the same height on both grids. On the fine grid it agrees to 0.15 % of the distance the shock travelled; on the coarse grid, 0.17 %. The criterion was 1 % and it was fixed before the run, not chosen afterwards.

Lead Researcher. The value of this problem is that it has an exact answer for one quantity and a hard, sharp, genuinely time-dependent structure for everything else. Agreeing with the exact part is the necessary condition; it does not by itself certify the rest of the picture.

Lead Engineer. The benchmark was priced at 20 core-minutes and cost 2.4. The estimate was the honest one available when it was written and it was high.

Caveats

The fine structure behind the main shock is shown here and is not among the quantities measured. The second shock and the wall jet beneath the main front are visible in the picture above. Read them as a picture, not as a number: no measured value for either is reported on this sheet.

This is an inviscid calculation. There is no turbulence model in it, so nothing here says anything about turbulence modelling.

The comparison is against exact theory, not against an experiment. No experimental measurement of this configuration exists to compare the rest of the structure against.